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**Perspective**

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# Barometric geolocators can reveal unprecedented details about the migratory ecology of small birds

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## ABSTRACT

Knowing the location of migratory birds throughout their annual cycle is fundamental for many questions in ornithology. Technological advances have provided several approaches, with increasing miniaturization allowing deployment on smaller and smaller birds. Here, we examine the strengths and limitations of barometric pressure geolocators (pressure tags) which have recently been shown to be extremely effective in tracking small birds (<25 g). The newly developed geolocation by pressure workflow (*GeoPressureR*) uses a global weather analysis dataset and hidden Markov movement model to reveal locations more accurately or continuously than can be achieved by other available light-weight devices (e.g., light-level geolocators, automated radio telemetry, and archival GPS tags). Additionally, pressure data can provide altitude information and document generally intractable flight behavior, as well as exact migratory phenologies. This new technology can be applied to important unresolved problems such as altitudinal migration, stopover site use, molt migration, vertical airspace use, and drivers of migratory flight initiation, duration, and direction. We provide an example of a pressure tag on a Swainson's Warbler (*Limnothlypis swainsonii*) deployed on its breeding territory in Virginia, USA, revealed the complete story of its migration to and from Cuba, including its refined wintering site, stopover sites, and migration routes, all with precise timing. Studies relying on pressure tags are still subject to biases related to recovery of the devices for data extraction and application to birds that exhibit extensive aerial movements (e.g., swallows and swifts) remains challenging. Widespread deployment of pressure tags could show unprecedented details of bird migration and other aspects of aerial behavior, which could benefit conservation and greatly enrich our understanding of avian movement ecology.

**Keywords:** atmospheric pressure, barometric geolocators, bird migration, flight behavior, nocturnal flight, phenology, stopover site

**How to Cite**

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## LAY SUMMARY

- Small migratory birds (<25 g) have limited options for high-precision tracking devices that can identify locations throughout their annual cycle.
- Barometric geolocators (pressure tags) archive atmospheric pressure, and with the use of the *GeoPressureR* analysis, can reveal migratory movements and stationary periods at high accuracies.
- We examine the strengths and limitations to using pressure tags to study the migration ecology of small birds.
- We discuss the range of potential questions that can be addressed with the implementation of pressure tags.
- We provide a proof-of-concept example of pressure data derived from a tracked Swainson's Warbler.
- Broad deployments of pressure tags can greatly improve annual cycle research and conservation.

## Los geolocalizadores barométricos pueden revelar detalles sin precedentes sobre la ecología migratoria de las aves pequeñas

## RESUMEN

Conocer la ubicación de las aves migratorias a lo largo de su ciclo anual es fundamental para muchas preguntas en ornitología. Los avances tecnológicos han proporcionado varios enfoques, con equipos cada vez más diminutos que permiten el despliegue en aves cada vez más pequeñas. Aquí, examinamos las fortalezas y limitaciones de los geolocalizadores de presión barométrica (marcadores de presión) que recientemente se han mostrado como extremadamente efectivos en el seguimiento de aves pequeñas (<25 g). La geolocalización por flujo de presión (GeoPressureR), recientemente desarrollada, utiliza un análisis de datos meteorológicos globales y un modelo de movimiento de

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Markov para revelar ubicaciones de manera más precisa o continua de lo que se puede lograr con otros dispositivos ligeros disponibles (e.g., geolocalizadores de nivel de luz, radio telemetría automatizada y marcas de GPS archivadas). Además, los datos de presión pueden proporcionar información de altitud y documentar un comportamiento de vuelo generalmente desconocido, así como fenologías migratorias exactas. Esta nueva tecnología puede aplicarse a problemas importantes no resueltos, como la migración altitudinal, el uso de sitios de parada, la migración durante la muda, el uso del espacio aéreo vertical y los impulsores del inicio, la duración y la dirección del vuelo migratorio. Proporcionamos un ejemplo de un marcador de presión colocado en *Limnothlypis swainsonii* en su territorio de cría en Virginia, EEUU, que reveló la historia completa de su migración hacia y desde Cuba, incluido su sitio detallado de invernada, los sitios de parada y las rutas de migración, todo con un detalle temporal preciso. Los estudios que dependen de los marcadores de presión aún están sujetos a sesgos relacionados con la recuperación de los dispositivos para la extracción de datos, mientras que su uso en aves que exhiben movimientos aéreos extensos (e.g., golondrinas y vencejos) sigue siendo un desafío. El despliegue generalizado de los marcadores de presión podría mostrar detalles sin precedentes de la migración de las aves y otros aspectos del comportamiento aéreo, lo que podría beneficiar la conservación y enriquecer enormemente nuestra comprensión de la ecología del movimiento aviar.

**Palabras clave:** comportamiento de vuelo, fenología, geolocalizadores barométricos, migración de aves, presión atmosférica, sitios de parada, vuelo nocturno

## INTRODUCTION

Tracking individual migratory birds throughout their annual cycle is paramount to describing migration routes, wintering and breeding locations, stopover sites, and phenology, thus revealing migratory behavior as well as the environmental and endogenous factors that influence decisions during migration (Marra et al. 2015). Since the first use of VHF transmitters in the 1960s, technological advances have provided increasingly accurate tools for determining birds' locations (McKinnon and Love 2018). Most recently, barometric pressure loggers have been added to archival multisensor geolocators (hereafter "pressure tags") for small birds (Liechti et al. 2018). Pressure tags record and store atmospheric pressure (Pa) at predetermined intervals (5–30 min), much like how a light-level geolocator records and stores light intensity data, and must be recovered to extract the archived data. Logged pressure data have already been used to document previously unknown flight behaviors in migratory species (Bowlin et al. 2015, Dhanjal-Adams et al. 2018, Meier et al. 2018, Liechti et al. 2018, Sjöberg et al. 2018, Briedis et al. 2020, Dreelin et al. 2018, Barras et al. 2021). Now, pressure data can be combined with global-scale weather reanalysis data and a movement model to map bird locations nearly continuously with great precision using the R package *GeoPressureR* (Nussbaumer et al. 2023a, 2023b), thus significantly amplifying the reach and use of its application (Lathouwers et al. 2022, Rime et al. 2023, Vīgants et al. 2023, Adamík et al. 2023). This technology and analysis overcome many limitations of standard approaches: location data are more accurate and complete than can be resolved from light-level geolocators; tags can be used on smaller birds (≥7 g) than costly archival GPS tags allow (>20 g); and spatial data can be collected beyond the reach of automated radio telemetry tags (e.g., nanotags and lifetags, hereafter "radio-tags") pinging Motus Wildlife Tracking System towers. Despite limited application, studies conducted so far demonstrate the potential for pressure tags to revolutionize how ornithologists can study small bird migration and add unprecedented detail not just for location estimation, but for flight behavior at fine temporal scales. In this paper, we examine the strengths and limitations of pressure tags for studying small bird migration and discuss the range of potential questions that can be addressed (Figure 2) across three main categories: Location estimation, phenology, and migratory behavior (Figure 1). We also present a proof-of-concept example of a recent study implementing pressure tags on a small passerine, the Swainson's Warbler (*Limnothlypis swainsonii*).

### Location Estimation

Accurately estimating unknown locations of a bird during periods of interest (typically wintering and migration) is the core component of geolocation and guides information such as migratory pathways and connectivity (McKinnon et. al 2013). The process to generate locations from pressure recordings in the *GeoPressureR (Nussbaumer and Gravey 2022)* reproducible workflow involves defining stationary periods corresponding to when the bird is assumed to be at the same location (at a 10- to 25-km resolution). This is done by identifying migratory periods based on drops in pressure (i.e., irregular shifts in the time series), which corresponds to increased altitude during migratory flight. For each stationary period, the pressure time series is extracted, cleaned, and matched with the global hourly weather reanalysis database ERA5 (Muñoz-Sabater et al. 2021) based on their spatiotemporal patterns. They are combined with a maximum-minimum elevation range calculated from altitude to generate a likelihood map (Figure 1A). Validation of pressure estimates to known locations have been found to be accurate to approximately 10–60 km, depending on the species' behavior and stationary duration (Nussbaumer et al. 2023a, G. Rhyne personal observation). As pressure recordings also allow for determining the flight duration between consecutive stationary periods, a hidden Markov model can be used to model the full migratory trajectory of the bird to produce (1) the most likely trajectory, (2) the marginal probability distribution map of each stationary period, and (3) simulated trajectories (Nussbaumer et al. 2023b). This movement model can be further improved by incorporating high-resolution wind speed information also from ERA5 to account for wind assistance and refine trajectories. See Nussbaumer et al. (2023a, 2023b) and the *GeoPressureR* Manual (https://raphaelnussbaumer.com/GeoPressureManual/) for a detailed explanation of the analysis.

The global coverage of *GeoPressureR* can now expand small bird tracking capabilities into regions lacking migration data, like much of the tropics where light-level data is overly broad from little day length variation and Motus receivers are absent or scarce. Pressure estimates include not only breeding and/or nonbreeding locations, but also stopover sites, which play a critical role in providing refueling and rest during stretches of migratory travel. An increasingly urbanized and developed world poses additional direct and indirect sources of mortality for migratory birds when traveling long distances (Nemes et al. 2023) and demands precise and complete stopover use data. Because pressure tags can estimate locations of < 1 day, the vast majority of stopover sites throughout a

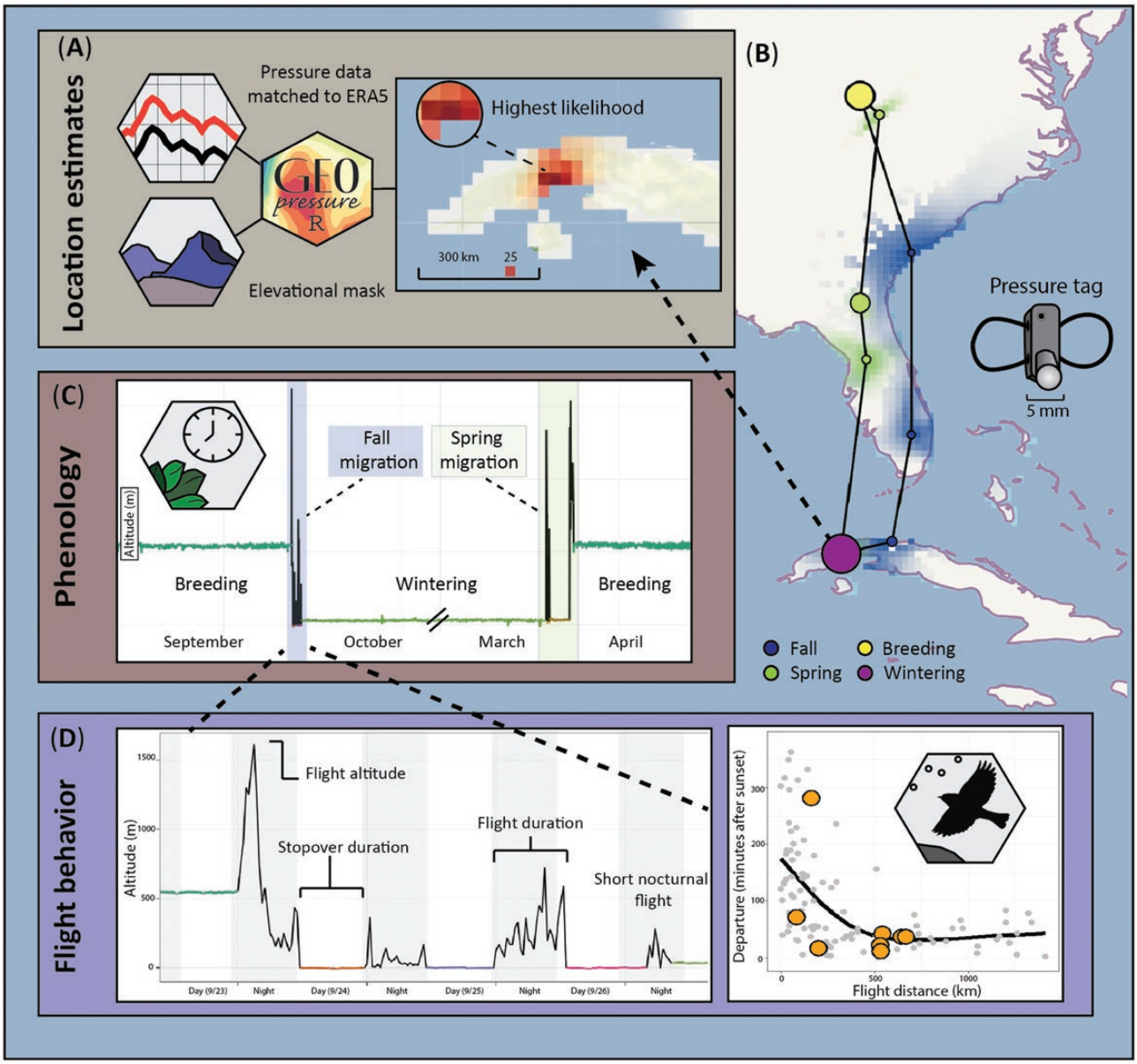



**FIGURE 1.** Migratory information derived from a single pressure tag deployed on a Swainson's Warbler (CB619) at Breaks Interstate Park, Virginia, USA. (**A**) Using the *GeoPressureR* package, pressure data are matched to a global weather reanalysis (ERA5) and combined with elevation to generate likelihood maps (25 × 25 km cells) as in this wintering location in western Cuba. The true location is either within or immediately surrounding the highest likelihood cell. (**B**) Continuous migratory trajectory with every stopover site. Point size indicates length of stay. Also shown is an illustration of the multisensor pressure tag (BARP50Z11-7, Migrate Technology) with a leg-loop harness. (**C**) A time series of altitude shows exact phenological events. (**D**) A closer look at 4 days of fall migration shows exact flight details and behavior, such as flight altitudinal patterns or short nocturnal flights (left). On the right, the relationship between departure time and flight distance can be examined for our example bird (larger orange points) and for our larger sample of Swainson's Warblers (smaller gray points).

bird's migration can be mapped. Error for short stopover (1–2 days) sites is typically broader given the limited temporal data, but still provide valuable information, whereas moderate to long stopover sites (>4 days) provide accurate estimates comparable to wintering locations. Stopover estimates can help resolve questions about selection of coastal or inland sites, or about relative use of stopover locations to map poorly understood flyways. For example, pressure data detailed impressive 15,000 km migrations of European Nightjars (*Caprimulgus europaeus*) along the Asian/East African flyway (Lathouwers et al. 2022). Breeding nightjars from Mongolia refueled at strategic stopovers to navigate multiple stretches of geographic barriers like the Gobi Desert, Arabian Peninsula, and Iranian Plateau, resulting in migration strategies distinct from western breeding populations (Lathouwers et al. 2022). Stopover use data like this can be invaluable for both conservation and studying migration energetics.

Continuous location estimates from pressure data can also help identify regions used during other life-cycle movements, particularly post-breeding movements or molt migration which are poorly understood. Vigants et al. (2023) identified post-breeding molting grounds of European Starlings (*Sturnus vulgaris*) hundreds of kilometers west of their breeding grounds in Latvia. Locations like these serve pivotal

roles during a bird's annual cycle and are important for assessing conservation strategies and understanding life history traits. Until now, researchers working with small birds had to extrapolate locations between fewer known points. By piecing each stationary period together, coupled with migratory flight durations and directionality, pressure data successfully maps an individual's complete annual cycle on a day-by-day basis.

Importantly, pressure geolocation estimation error is distributed irregularly across space, most notably following elevational gradients. Altitudinal measurements make pressure tags particularly suitable to study elevational movements at seasonal or daily scales and distinguish topographical and ecological features that correspond to different habitats (i.e., montane or lowland). Accounting for the natural variation of pressure in *GeoPressureR*, we can refine the vertical position of the bird and, consequentially, describe altitudinal movement which were previously within the uncertainty of the vertical position (precision of ~10 m instead of ~100 m previously). By distinguishing transition periods in close spatial proximity, pressure tags are well fit for tracking altitudinal migrants, which are globally understudied, that utilize different elevations throughout the year but may not travel significant horizontal distances. Rime et al. (2023) demonstrated this utility by finding Northern Wheatears (*Oenanthe oenanthe*) to stay at lower elevations after arriving at montane breeding regions in the Swiss Alps until conditions at higher elevations were favorable for nesting. More frequent altitudinal shifts do require more effort and attention within *GeoPressureR* to distinguish different altitudes at the same location. However, this movement data can describe space use of different elevations over time, which may serve as a proxy for resource availability, territory size, or other behaviors (Barras et al. 2021). These distinctions are realized by continuously measuring altitude, defining a transition period, and using the increase in data points to discern close locations.

## Phenology

The timing of migration departure and arrival to both breeding and wintering grounds has a profound impact on resource availability, territory acquisition, mate selection, and overall fitness (Spottiswoode et al. 2006, Gienapp and Bregnballe 2012). Climate change has caused shifts in seasonal resource phenology, notably advancing spring "green-up" in the northern hemisphere, affecting migration phenology of many species and posing threats to long-distance migrants (Marra et al. 2005, Youngflesh et al. 2021, Connare and Islam 2022). To properly assess intra-specific and seasonal differences of migration schedules, it is imperative that tracking technologies provide precise dates of migration activity. Current methods often provide incomplete timelines or have difficulties delineating transitions with high accuracy. Because pressure tags reveal migratory activity that separate defined locations, they can produce an exact schedule from staging to departure to stopovers to arrival (Figure 1C). Pressure tag data shows how tracked birds distribute time and energy across the duration of migration (i.e., the ratio of time spent refueling at stopover sites versus time in flight; Briedis et al. 2020), as well as seasonal or population differences in migratory durations. Length of stay is also determined at each stopover region and can serve as an index of habitat quality, helping differentiate long-term refueling from short-term stops (Herbert et al. 2022). This level of detail would greatly improve long-term phenological studies, especially in a changing climate.

## Flight Behavior

Adding more precise points in space means pressure tags will add greater detail for standard questions about general patterns of bird migration. Perhaps even more exciting is the potential to know much more about in-flight behavior. Visualization of flights, especially nocturnal flights, by small- to medium-sized migrants is arguably the most novel and compelling aspect of pressure tags. Sudden deviations in pressure indicate flights previously difficult to quantify, including departure and arrival times of each individual flight, overall flight duration, average and maximum flight altitudes, and more (Figure 1D; Liechti et al. 2018, Sjöberg et al. 2018, Briedis et al. 2020, Sjöberg et al. 2021, Adamík et al. 2023). We can also better understand drivers and speed metrics of migratory flight, as an important outcome of the movement model is to extract the groundspeed, airspeed, and windspeed in vectorial form (direction and magnitude) for each individual flight (Nussbaumer et al. 2023b).

For decades, researchers have sought to understand the complex mechanisms of nocturnal flight in small birds and evaluate extrinsic cues that influence initiation and landing, such as sunset, temperature, wind, cloud cover, air pressure, and precipitation, as well as intrinsic factors such as fuel load, age, and sex (Åkesson and Hedenström 2000, Burnside et al. 2021, Cooper et al. 2023a, 2023b, Rüppel et al. 2023). Studies that investigate these mechanisms in small birds typically track birds locally as they depart from a single location but require assumptions about the individual's flight and next destination. Weather radar has also been useful for quantifying large-scale migratory movements but cannot track individual birds across night and day and over multiple days. Now, pressure data can help answer these questions by providing empirical comparisons of migratory departure and local environmental conditions, and match each flight with distance, duration, altitude, and destination across different species. Pressure tag data can be combined with local environmental data to examine mechanisms underlying this variation and help explain decisions birds make based on previous days' flights and stopovers.

By plotting flight altitudes across two known locations, we gain valuable insight into how birds manage ecological barriers, such as large bodies of water, mountain ranges, urban areas, or resource-poor areas like deserts, which are energetically expensive to cross (Deppe et al. 2015). Sjöberg et al. (2021) tracked Great Reed Warblers (*Acrocephalus arundinaceus*) with pressure tags and found unexpected patterns when analyzing prolonged flights across geographic barriers like the Sahara Desert and Mediterranean Sea. As these long flights transitioned from nocturnal to diurnal, birds rapidly climbed from nighttime cruising altitudes of ~2,300 m to > 5,000 m throughout the day, surpassing previous perceptions of altitudinal limits for passerines (Sjöberg et al. 2021). Liechti et al. (2018) observed dramatic altitude changes throughout migratory flights, presumably from birds adjusting to environmental conditions most favorable for flight. We know qualitatively that wind significantly impacts migratory flight efficiency, orientation, and altitude (Alerstam 2011, Shamoun-Baranes et al. 2017). Now, behavioral pressure data, combined with fine-scale wind speed data during flight, can help researchers better understand

influences of wind and how birds search for optimal atmospheric conditions in flight (Werfeli et al. 2022). Flight space-use across the landscape may also be useful in conservation planning to avoid infrastructure collisions with wind turbines and buildings (Nemes et al. 2023).

Pressure tags can also reveal other non-migratory behaviors, as demonstrated by several studies prior to the development of *GeoPressureR*. Twilight ascents by aerial insectivores were thought to be related to migratory orientation until pressure-tagged Alpine Swifts (*Apus melba*) displayed this behavior year-round, perhaps for social purposes (Meier et al. 2018). Dhanjal et al. (2018) used pressure tags to identify and describe social-group dynamics of European Bee-eaters (*Merops apiaster*) throughout their annual cycle. Dreelin et al. (2018) found differences in atmospheric foraging strata between swallow species. Many other possible studies relating to daily routines or breeding activity may be revealed, such as aerial breeding display behaviors or roosting cycles. The deployment of pressure tags across space, time, and taxa can greatly improve the ability to track individual activity that were previously challenging to observe across seasons.

## Strengths and Limitations

Pressure tags possess unique data characteristics that make them preferable alternatives to light-level geolocators, radio-tags, and GPS tags, but in different ways. Location estimation from barometric pressure is advantageous over light-based estimation from geolocators for four important reasons. First, temporal resolution of light data is limited to two daily positions from twilight events (sunrise and sunset), whereas pressure data are compared to ERA5 hourly (24 points per day). Second, the precision of location estimates does not rely on calibration, contrary to light-level calibration procedure that essentially assumes constant shading conditions throughout the annual cycle. Third, spring and autumn equinoxes, periods that often correspond with bird migration, do not affect pressure data, unlike light-level tags which cannot estimate latitude when day length is equal across latitudes. Similar light estimation issues regarding day lengths arise for birds at both high latitudes and the equator. Fourth, pressure tags are not nearly as sensitive to environmental errors that compromise light data, including shading from habitat, topography, bird behavior, weather, cloud cover, and feather placement (Lisovski et al. 2012, Fudickar et al. 2012). This shading interferes with delineation of twilight periods, and thus impedes latitude and longitude estimates for that day. Shading can be especially problematic for species that occupy dark habitats like forests. In contrast, geopositioning by pressure utilizes all environmental and climatic fluctuations in pressure appearing in both the recorded data and ERA5 reanalysis data to improve the precision of the location estimate (Nussbaumer et al. 2023a)). Although advanced tools have allowed for the construction of general migratory pathways from light data (Sumner et al. 2009, Rakhimberdiev et al. 2017), light data cannot reliably identify exact migratory transitions nor estimate short-duration locations like stopovers (though it may provide complementary longitudinal data for stopovers from pressure) and can even produce misleading results (Lisovski et al. 2018). Pressure data removes movement assumptions by delineating transitions and employing an inherently localized dataset for a more rigorous analytical approach to provide continuous, reliable, and accurate locations.

Radio-tags and archival GPS tags can provide accurate location data but cannot match the number of points in space and time obtained by pressure tags. Radio-tags provide the important advantage of collecting relatively accurate data (~20 km range) without the need of recovering the device (Taylor et al. 2017). However, this comes at the cost of erecting and maintaining extensive networks of receiver towers. This is typically expensive and difficult in remote but important regions for migratory birds (i.e., Afrotropics, Amazonia, southeast Asian islands, high arctic, etc.) or in regions that experience annual extreme weather events like the U.S. coast of the Gulf of Mexico. Although it is rapidly expanding, this network currently only provides quality coverage in the U.S. and not across the majority of the global landscape (www.motus.org). In addition, because of these network limitations, radio-tag data cannot provide continuous locations throughout the birds' annual cycle, but rather brief timestamps when birds pass through an area with tower coverage. Radio-tags are also susceptible to topography, vegetation, and radio wave interference, and some can produce false-positive detections from other tags. Locations from archival GPS tags are the most accurate of available technologies, but weight limitations greatly constrain both the size of species that can carry tags and the number of points that can be taken throughout deployment (Iverson et al. 2023a). Species > 30 g can obtain 30–100 pre-programmed locations throughout deployment (Musseau et al. 2021, Iverson et al. 2023b), whereas the smallest species capable of carrying GPS tags (>20 g) may only accumulate a dozen or so points (Hallworth and Marra 2015), thus sacrificing continuous location data for fewer fine-scale locations. Satellite position transmissions are also susceptible to interference from vegetation and poor satellite coverage in remote regions, which may be crucial for tagged birds scheduled for limited points. Additionally, archival GPS tags are more expensive than geolocators or radio-tags. These technologies have proven to be useful tools for studying migration, but may sometimes be less effective than pressure data at reconstructing the full trajectory of a small bird continuously over time.

Despite their advantages, there are 3 primary limitations to using pressure tags for tracking. First, tags are archival, so they must be recovered from a recaptured bird. Deployment and recovery of pressure tags are best suited for species for which individuals can be reliably encountered. Consequently, data are limited to birds that survive and return, rather than an unbiased sample from the population. This underlies the importance of data from non-archival methods (automated radio telemetry) to reduce data biases, especially for female and juvenile birds which are generally difficult to recapture. Pressure and Motus data can potentially be quite complementary, and ideally could be combined within a single tag for maximum data collection. Second, location and migration phenology estimation does not apply well to aerial feeders (e.g., swallows, swifts, etc.) that regularly fly at varying altitudes due to the difficult nature of distinguishing daily movements and migratory movements. However, even a few hours of roosting of these aerial feeders can be sufficient for positioning. Third, to be accurate, geoposition by pressure can require a significant effort of data labeling to identify migratory flight and distinguish natural variation of pressure from a bird's vertical movement, depending on the species' behavior and complexity of their migration. Labelling can be tedious and sometimes subjective. However, the *GeoPressure* manual offers guidelines with examples and tools to make this exercise easier and provides standards to make the analysis reproducible and

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|---|---|---|---|
| Location Estimation | **Migratory trajectories** | Migratory connectivity across populations<br>Environmental conditions on nonbreeding grounds<br>Migration patterns (e.g., loop migration)<br>Pre- and post-breeding movements<br>Molt migration | *9, 11, 12, 13* |
| | **Stopover sites** | Shared stopover sites<br>Length of stay at stopover sites (rest stop vs. refuel)<br>Coastal vs inland stopover use<br>Site use prior to and after crossing barriers | *9, 11, 12, 13* |
| | **Elevational movements** | Elevational differences between breeding / nonbreeding<br>Altitudinal migrant ecology<br>Daily and seasonal elevational movements | *7, 12* |
| Phenology | **Phenology** | Migratory timeline shifts across years<br>Seasonal differences in migration duration and speed<br>Stopover to flight ratio<br>Population differences in migration timing | *1, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13* |
| Flight Behavior | **Migratory flight** | Flight durations<br>Flight speeds (airspeed, groundspeed)<br>Flight initiation and landing times | *1, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13* |
| | **Flight altitudes** | Average, minimum, and maximum altitudes<br>Altitudes when crossing barriers<br>Altitudinal fluctuations<br>Ascent and descent rates | *5, 6, 7, 8, 11, 12, 13* |
| | **Weather** | Weather influence on flight initiation, duration<br>Local movements due to large weather events | *5, 10* |
| | **Wind** | Wind drift on migratory routes<br>Strength and directionality on migratory departure<br>Wind support during flight<br>Adjustment to wind shear | *10* |
| Other | **Non-migratory flight** | Aerial foraging strata and timing<br>Aerial breeding displays | *2, 3* |
| | **Other** | Roost sites<br>Bat migration ecology | |

**FIGURE 2.** Topics about full annual cycle location, migration phenology, and flight behavior for small birds that can be answered or improved upon by using pressure tags. The rightmost column provides numbers corresponding to currently published articles that have used pressure data to address the preceding topics: (1) Dhanjal et al. (2018), (2) Dreelin et al. (2018), (3) Meier et al. (2018), (4) Leichti et al. (2018), (5) Sjöberg et al. (2018), (6) Briedis et al. (2020), (7) Barras et al. (2021), (8) Sjöberg et al. (2021), (9) Lathouwers et al. (2022), (10) Werfeli et al. (2022), (11) Adamík et al. (2023), (12) Rime et al. (2023), and (13) Vigants et al. (2023).

verifiable. The labor of identifying migratory flight can also be alleviated with accelerometer data from the same device to better delineate constant migratory movements. As this analysis continues to be implemented, we emphasize the importance of users publishing publicly-accessible data and code to ensure reproducibility and transparency.

## Proof of Concept

To demonstrate the efficacy and advantages of pressure tags, especially for forest birds, we highlight results of a single tag (CB619, deployed in Virginia, USA) from a range-wide study on Swainson's Warblers (Rhyne et al. personal observation). We deployed multi-sensor tags that recorded both pressure and light (BARP50z11-7, 0.58 g, Migrate Technology, UK) to evaluate migratory connectivity of this secretive species across the southeastern U.S. in 2021–2022. When analyzing light-level data from this tag, we encountered poor-quality light data that led to uncertain and overly broad (+200 km) wintering location estimates and little information on migratory pathways and phenology (Supplementary Material Figure 1). Instead, we implemented the *GeoPressureR* v.2.7 analysis to pressure data, independent of light data, and successfully gained (1) greater wintering location precision of > 5x, (2) continuous migratory trajectories complete with stopover sites, (3) a detailed timeline of migration phenology, and (3) detailed insight into flight behavior (see Data availability statement in the Acknowledgments).

This tagged bird wintered south of Havannah, Cuba, with stopovers throughout Florida (Figure 1). Although the

likelihood distribution maps may superficially appear broad, our ground-truthing to known locations (deployment sites) found an average error of 26 km from the most likely location estimate. This means that the bird's true location, especially for long-term wintering estimates, is likely either within or immediately surrounding the grid cell (25 × 25 km) of the highest likelihood (i.e., the darkest cell; Figure 1A). Shorter stopover sites (Figure 1B, smaller points) have greater uncertainty given the reduced datapoints but are still reliable and informative. Pressure data uncovered an important stopover region used by several other tagged birds, the Okefenokee Swamp region along the southern Georgia border. This full trajectory also provided an exact migratory timeline that detailed a shorter post-breeding migration (24–27 September) than a longer pre-breeding migration (6–15 April) with more time spent at stopover sites (Figure 1C). Additionally, pressure data revealed novel flight behaviors based on a larger sample of Swainson's Warblers ($n$ = 14, see GitHub page). For example, we observed nonlinear relationship in departure times relative to sunset, with long-distance flights associated with earlier departures (<1 hr) after sunset but more variable departure times, up to 4 hr after dusk, for short-distance flights (Figure 1D, right panel). We observed rapid fluctuations, >1,000 m in < 30 min, over the course of long flights, possibly demonstrating the birds searching vertically for favorable conditions. Lastly, we commonly noted unique short nocturnal flights (<2 hr) during spring migration, perhaps associated with birds refining their search for their previous breeding territory as they drew closer.

## Conclusions

Pressure tags present a more accurate, data-rich, and affordable geolocation alternative for small birds that, importantly, can be deployed globally. However, given the logistic constraint of recovering archival tags, researchers must clarify their objectives and questions to decide if pressure tags will meet their personal requirements. Several manufacturers offer barometric loggers at similar prices to light-level geolocators and developers are continuing to improve their design (Brown et al. 2023). Currently, there are multisensor geolocators as small as ~0.35 g that are regularly supplemented with additional sensors like accelerometers, light, and temperature loggers. We also recognize the potential for pressure tags to be used on bats, which face even more complex tracking challenges. Many sub-fields of ornithology would greatly benefit from implementing pressure tags to answer questions regarding energetics and fuel storage, migratory optimization, stopover habitat use, connectivity, behavior, molt, and much more. We encourage the deployment of pressure tags to uncover the natural history of migratory behaviors and advance basic ornithology and bird conservation.

## Supplementary material

Supplementary material is available at *Ornithology* online.

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## Ethics statement

Required permits were obtained to capture, handle, and tag birds for data collection.

## Conflict of interest statement

The authors have no competing interests.

## Author contributions

G.S.R. and P.C.S. conceived the idea. G.S.R., R.N., and M.B. collected and analyzed data. R.N. and M.B. developed the methods used. G.S.R. and P.C.S. wrote the paper. All authors contributed substantial materials.

## Data availability

All raw and analyzed data are publicly available and reproducible on the Github repository (https://github.com/grhyne/SWWA_Pressure), as well as for the GeoPressureR package (https://github.com/Rafnuss/GeoPressureR)

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